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SmartPresence: Wi-Fi-based online attendance management for smart academic assistance

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Abstract

SmartPresence, a Wi-Fi-based attendance management system marks a significant leap in evaluating student performance, replacing traditional manual methods with pen-and-paper timesheets or paper-based student signatures. This innovative system utilizes smartphones and Wi-Fi signals, allowing students to connect their devices to a designated router upon entering the classroom for attendance registration. The captured data is stored centrally, instantly notifying both administrators and teachers in real time. This approach not only revolutionizes attendance tracking but also enhances overall classroom management by leveraging Wi-Fi-enabled devices like smartphones and laptops. This system introduces a progressive solution, utilizing contemporary connectivity for a seamless and intelligent attendance monitoring experience for educators and students alike.

Keywords: Smart attendance, Wi-Fi attendance, IoT, Smart classroom, Online attendance, Academic automation

Introduction

One of the key factors influencing students' academic performance is class attendance [1]. The landscape of attendance management within the academic sector has undergone a significant transformation, moving from traditional manual methods to sophisticated web and software-based systems, and now, embracing cutting-edge IoT and Wi-Fi-based attendance systems. The conventional manual attendance system, once ubiquitous in educational institutions, entailed time-consuming processes, often involving the calling of names and paper-based record-keeping. As technology advanced, the shortcomings of these manual systems became increasingly apparent, prompting the need for more efficient and automated solutions. The evolution of attendance systems in academia has seen a paradigm shift with the advent of web and software-based solutions. These systems introduced digital interfaces, enabling streamlined attendance tracking and management. Educators and administrators could now leverage technology to simplify the attendance recording process and access attendance data more conveniently. However, as the educational landscape continued to evolve, the demand for even more advanced and real-time solutions prompted the integration of IoT and Wi-Fi technologies into

attendance management. IoT-based attendance systems represent a notable leap forward, leveraging interconnected devices to provide a more comprehensive and automated approach. These systems often involve smart devices or wearables that students use to mark their attendance, eliminating the need for manual input. The interconnected nature of these devices ensures that attendance data is captured in real time, allowing for immediate insights into attendance patterns. While IoT systems addressed several limitations of manual and basic digital attendance systems, the quest for even greater efficiency and adaptability led to the emergence of Wi-Fi-based attendance systems.

Wi-Fi-based attendance systems stand at the forefront of modern attendance-tracking solutions in education. These systems utilize Wi-Fi signals and mobile technology to offer real-time attendance-tracking capabilities. The seamless integration with existing infrastructure ensures scalability across diverse educational settings. The significance of Wi-Fi-based systems lies in their ability to automate attendance processes, reduce administrative burdens, and provide educators with immediate insights. These systems not only revolutionize attendance tracking within educational institutions but also hold the potential for adaptation in other sectors, showcasing their versatility and broader implications for efficient data management. As we delve into the intricacies of these attendance systems, this exploration aims to dissect the evolution from manual methods to sophisticated Wi-Fi-based solutions. It delves into the challenges posed by manual attendance systems, the transformative impact of web and software-based solutions, the enhanced capabilities offered by IoT integration, and ultimately, the technological prowess of Wi-Fi-based attendance systems. Understanding this progression is crucial for appreciating the strides made in attendance management and anticipating the future trajectory of these systems as they continue to redefine efficiency and transparency in educational institutions.

We commenced by providing an all-inclusive literature review and contribution to knowledge in Sect. "Literature review". We provide the methodology in Sect. "Methodology", which involves a system architecture overview, flowchart, IoT device design, and integration. Sect. "Implementation and discussion", discusses our developed software and practical usage in the academic field. The concluding part of the work is described in Sect. "Conclusion".

Literature review

Traditional methods of monitoring attendance face limitations hindering effectiveness. Manual processes are prone to human errors, are time-consuming, and divert resources from critical tasks. Card-based systems are susceptible to proxy attendance. A technological intervention is needed to streamline attendance management, enhance accuracy, and improve security. Leveraging Wi-Fi signals in educational and workplace settings presents an opportunity for a sophisticated and automated system. Wi-Fi signals are widespread, and existing infrastructure makes it a promising option. This approach allows accurate attendance tracking without manual input or physical cards. Wi-Fi integration offers advantages, utilizing prevalent devices like smartphones and laptops for convenient and real-time attendance capture, reducing errors associated with delays or manual recording. Several renowned attendance management systems have been used frequently. Some of them are:

Attendance Management System (AMS)- AMS is crucial for tracking student attendance in educational institutions. It efficiently manages student attendance details for specific classes, emphasizing knowledge enhancement through ongoing software development [2].

Paper-based to Smartphone Attendance- Historically, manual attendance processes involved time-consuming name-calling or paper-based sign-ins. The introduction of smartphone-based attendance not only saves time and paper but also reduces energy consumption, marking a sustainable shift [3].

Challenges of Existing Systems- Previous attendance automation systems faced challenges, lacking Graphical User Interface (GUI), automation for caretaker notification, or being desktop-dependent. The need for a mobile-based system is emphasized, offering energy efficiency and wider accessibility [4].

Android-Based Student Attendance Management- In the evolving landscape of technology in Indian education, the Android-based Student Attendance Management System emerges as a pivotal tool. It provides daily attendance maintenance, empowers teachers with mobile attendance tracking, and offers real-time status access for authorized personnel [5].

Biometric Solutions for Attendance- Biometric methods, especially fingerprint recognition, enhance attendance accuracy. Microcontrollers with fingerprint sensors transfer attendance data to computers via USB or SD card. Integration with communication modules like GSM, Wi-Fi, or LAN facilitates data access through mobile applications [6, 7].

Wi-Fi-Based Attendance Tracking- Utilizing Wi-Fi signals for attendance via mobile devices is proposed. Wi-Fi signal strength and MAC address detection authenticate student presence without connecting to access points, eliminating user limits [8].

Facial Recognition System- An alternative proposal suggests a facial recognition prototype using neural network algorithms. This near-real-time system verifies users for high-level security information systems [9].

Green and Mobile-Friendly Attendance- Mobile-based attendance reduces energy consumption, eliminates paper wastage, and provides instant status updates for parents. The ubiquity of mobile devices makes it an eco-friendly and efficient solution [8, 10].

Smart Attendance Application- Integrating GPS technology into attendance systems ensures an intelligent and location-specific approach. Teachers set up attendance fences, students use Android phones for fingerprint attendance, and cloud-based recording streamlines the process. Automated notifications to parents enhance communication and reduce manual record-keeping [11].

While several attendance monitoring systems like biometrics, fingerprint scanners, and iris scanners exist in the market, they often incur high costs for the implementation and maintenance of a comprehensive student database. Ramakrishnan et al. [12] developed an application that utilizes Wi-Fi and location services to record attendance on campus. Another authors [13] developed a IoT-based attendance system using Amazon Web Services. Gujarkar et al. [14] developed a high efficiency IoT-based attendance system for industry purpose. Anoliefo et al. [15] developed seamless real-time class attendance management system using Long Range (LoRa) and Wi-Fi handshake.

This study introduces an innovative approach using Android smartphones with Wi-Fi capability for student attendance tracking. The system captures attendance data every hour, storing it in a database for easy access and generation of student attendance reports. This method offers an efficient and cost-effective solution compared to traditional systems in the market. In our SmartPresence system, the development undertook a comprehensive exploration focused on designing a sophisticated system architecture that effectively incorporates Wi-Fi signal tracking into the established infrastructure of colleges and universities. The paper's contribution to knowledge includes:

- *Automation for Efficiency*- Our system automates the labor-intensive process of manual attendance tracking, replacing traditional methods with a streamlined, technology-driven approach. This significantly reduces the time and effort required by educators and administrators to manage attendance records.
- *Real-time Tracking with Wi-Fi Integration*- Leveraging the power of Wi-Fi signals and mobile technology, our system introduces real-time tracking of student attendance. This innovation allows for instantaneous data capture, providing educators with accurate and up-to-date information for timely interventions and decision-making.
- *Resource Optimization and Future Integration*- By eliminating the need for manual data entry and paper-based record-keeping, our system optimizes resources, including time, manpower, and paper. Additionally, the system's design allows for future enhancements, such as IoT integration, real-time data analysis, and blockchain technology, ensuring its adaptability to evolving technological trends.

Methodology

This paper considers two crucial workflow steps to actualize the research. The steps involved in our system are: collecting & analyzing data, designing and implementing IoT devices, developing WebApp for management, and evaluating the system in a practical field. The full system architecture overview is illustrated in Fig. 1. Table 1 represents the summarized meta-information of our developed system.

Data collection & analysis

The process of data collection involved two primary elements:

- i. *Analysis of Wi-Fi Signals*: This involved collecting data on the fluctuations in Wi-Fi signal strength in various areas within the educational institution, ensuring precise tracking.
- ii. *Soliciting User Feedback*: Gathering input from both administrators and students concerning the usability and acceptance of the Smart Attendance System.

IoT device implementation

In our SmartPresence system, the development undertook a comprehensive exploration focused on designing a sophisticated system architecture that effectively

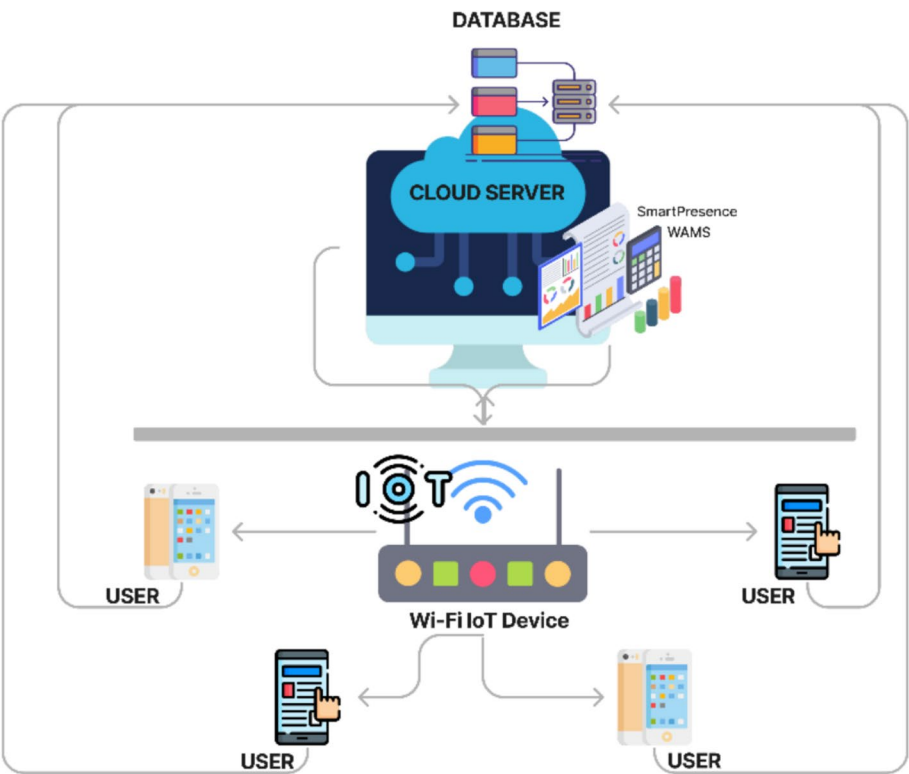


Fig. 1 System architecture

Table 1 Metadata information

System metadata information title	Information
Present code version	v1
Permanent link of WebApp	https://github.com/abrahametry/SmartWifiAttendance
Legal code license	MIT License
Versioning system	Git
Tools & technologies used	HTML, CSS, JS, PHP, C + +, VSCode, Arduino IDE, Bootstrap, ESP32

incorporates Wi-Fi signal tracking into the established infrastructure of colleges and universities which is shown in Fig. 1. This intricate design aimed at seamlessly integrating various elements, encompassing both hardware and software components, as well as defining the communication protocols necessary for smooth operation.

In terms of hardware, the development delved into the specification and inclusion of Wi-Fi-enabled devices as essential components of the proposed system. These devices play a crucial role in capturing and transmitting Wi-Fi signals for attendance-tracking purposes. Figures 2 and 3 depict the IoT device block diagram and connection, respectively. Table 2 represents the ESP32 pin reference. At a glance, the IoT device connection is as follows:

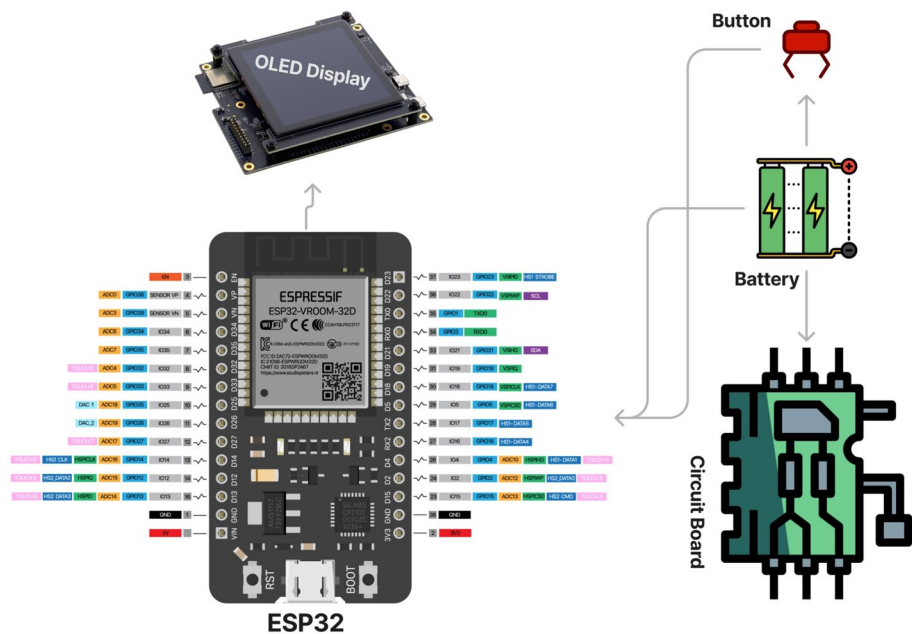


Fig. 2 IoT device connection block diagram

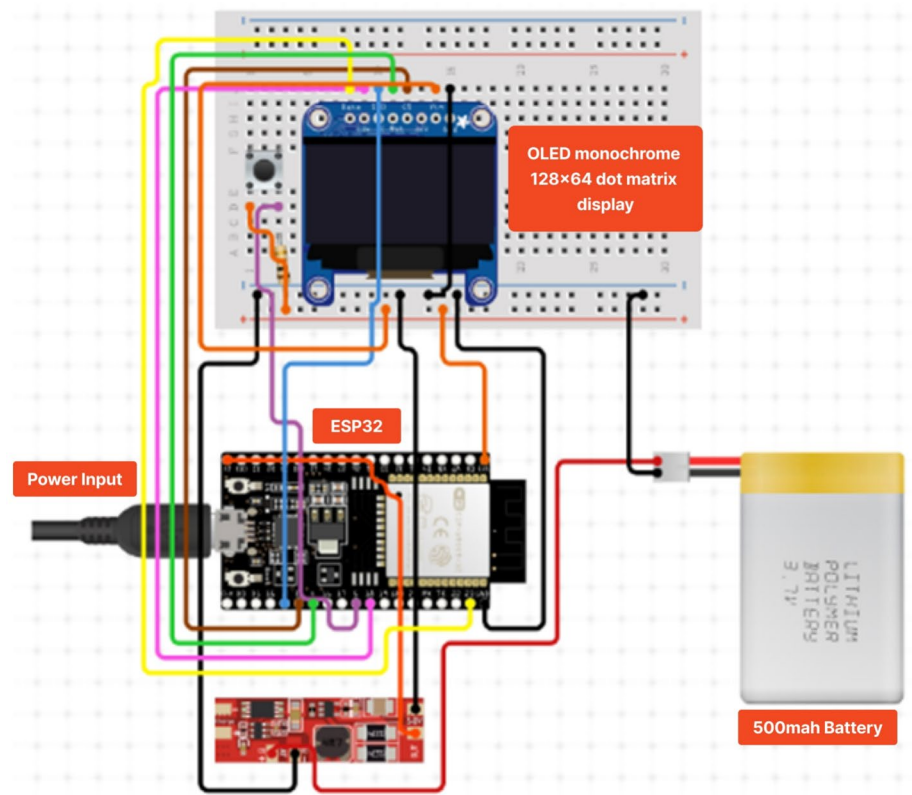


Fig. 3 IoT device connection

Table 2 ESP32 pin reference details

GPIO Pin	Pin on ESP32	Information
0	–	Pulled HIGH. Connected to BOOT Button
1	TX0	Do not use while TXing
2	YES	Pulled LOW
3	RX0	Do not use while RXing
4	D4	Pulled LOW
5	D5	Pulled HIGH
6	–	Connected to SPI Flash IC
7	–	Connected to SPI Flash IC
8	–	Connected to SPI Flash IC
9	–	Connected to SPI Flash IC
10	–	Connected to SPI Flash IC
11	–	Connected to SPI Flash IC
12	D12	Pulled LOW. Boot fails if pulled HIGH as it sets voltage of internal voltage regulator
13	D13	
14	D14	
15	D15	Pulled HIGH
16	RX2	UART2 RX
17	TX2	UART2 TX
18	D18	
19	D19	
21	D21	I2C SDA
22	D22	I2C SCL
23	D23	
25	D25	
26	D26	
27	D27	
32	D32	
33	D33	
34	D34	Digital Input Only. No Digital Output
35	D35	Digital Input Only. No Digital Output
36	YES	Digital Input Only. No Digital Output
39	YES	Digital Input Only. No Digital Output

ESP32 Display:

- i. SDA (Data Line): Connected to the ESP32's default SDA pin (Pin 21).
- ii. SCL (Clock Line): Connected to the ESP32's default SCL pin (Pin 22).
- iii. Power: Connected to ESP32's 3.3 V pin for power.
- iv. Ground: Connected to ESP32's GND pin.

ESP32 to Button:

- i. Button Pin: Connected to GPIO 35 on the ESP32.
- ii. Power: Connected to ESP32's 3.3 V pin for power.
- iii. Ground: Connected to ESP32's GND pin.

ESP32 to Wi-Fi Manager:

The Wi-Fi Manager library is used to facilitate the configuration of Wi-Fi settings. It doesn't require specific hardware connections, but it may use the default ESP32 GPIO pins for configuration.

Web application development

On the web-system front, the development focused on the design of databases to store attendance-related information securely and efficiently. Additionally, the development of application interfaces, which serve as user-friendly interaction points, was a pivotal aspect of the software architecture. These interfaces facilitate the input, retrieval, and management of attendance data by administrators, educators, and other stakeholders.

The initial phase of this system development process utilized the Wi-Fi module's hotspot feature. Afterward, the student's individual information became visible on the Node device, concluding with the transmission of recorded data to the server. Conversely, in the context of lecture attendance, the ESP32-based NodeMCU associated with the hotspot module connecting to the Class Node will employ a Wi-Fi or internet connection to forward the student attendance data to the server, facilitating the smart application's access to student attendance information. The overall System process flowchart is depicted in Fig. 4.

Implementation and discussion

Figure 5 illustrates the overall system features. This attendance system operates through a well-structured multi-stage process designed for the efficient capture and management of student attendance data. In the initial stage, authorized personnel, such as teachers or administrators, carry out a meticulous registration process. This involves the completion of a detailed registration form, followed by a comprehensive student data scanning procedure facilitated by assigned operators. The teacher ensures the accurate and complete entry of crucial student information, including ID, intake, section, and number. Once this data is validated, it is securely transmitted to the server for storage in the database.

The second stage of the process begins when students enter the classroom for a lecture. To begin the attendance process, students use designated hotspots within the Class Node, supervised by the respective lecturer. This interaction generates attendance data, promptly transmitted from the Class Node to the central database for secure storage on the server. Subsequent lectures, whether within the same class or across different courses, follow a consistent attendance protocol. Students engage with the Class Node hotspots under the supervision of the teacher, triggering the transmission of attendance data to the server for comprehensive record-keeping. This systematic approach ensures the accuracy and integrity of attendance records throughout various courses and classes.

Attendance Management System has two main modules for proper functioning:

1. The teacher module has the right to create any new entry of student details.
2. Users have the right to make daily attendance-generating reports. Attendance reports can be taken by giving details of students' details, dates, and classes

Figure 6 demonstrates some example images of interface of our developed system.

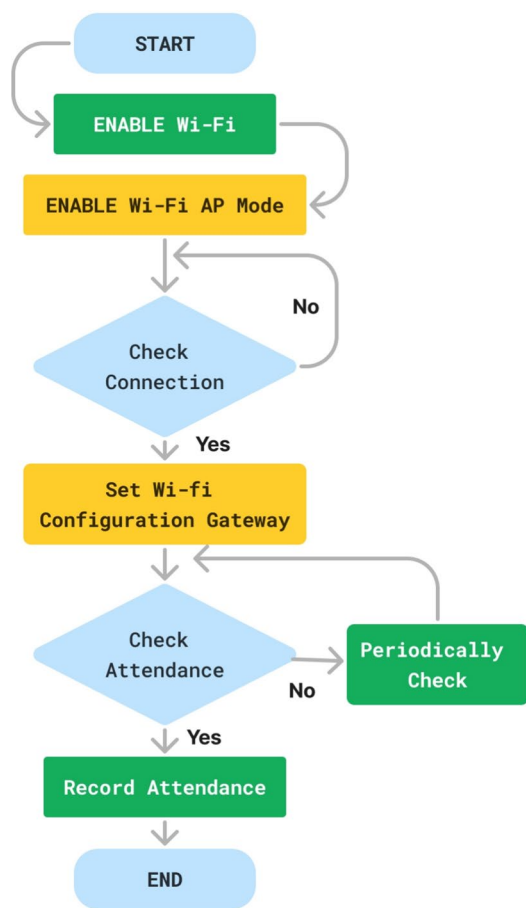


Fig. 4 System process flowchart

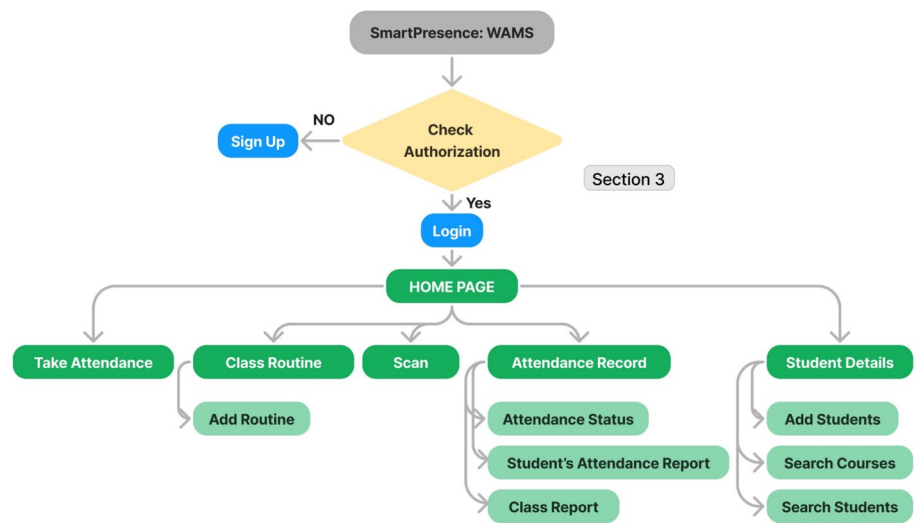
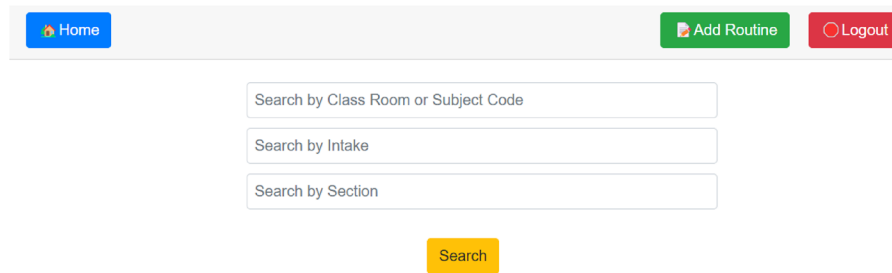


Fig. 5 System features flowchart

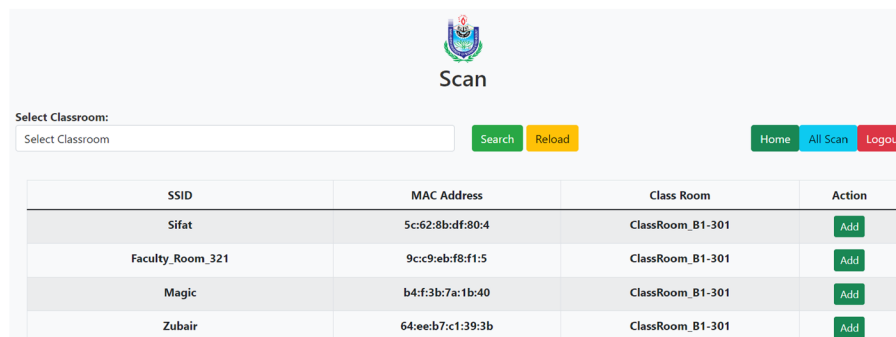
This SmartPresence: Wi-Fi-based Attendance Management System (WAMS) presents a comprehensive array of features that collectively contribute to its efficiency and user-friendliness. The “Take Attendance” module streamlines the attendance-tracking process with input fields for room number and course, a search button for retrieval, and a table displaying student information with buttons for marking attendance (present/absent). The inclusion of a bulk submission button enhances convenience for educators.



Sunday Routine

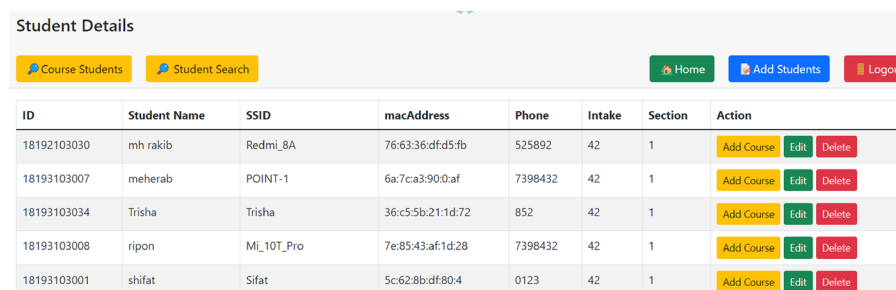
Start Time	End Time	Subject Code	Intake	Section	Class Room	Action
08:00:00	09:00:00	CSE 205			ClassRoom_B1-301	Delete
16:30:00	17:30:00	CS100	42	1	ClassRoom_B1-301	Delete

(a)



SSID	MAC Address	Class Room	Action
Sifat	5c:62:8b:df:80:4	ClassRoom_B1-301	Add
Faculty_Room_321	9c:c9:eb:f8:f1:5	ClassRoom_B1-301	Add
Magic	b4:f3:b7:a:1b:40	ClassRoom_B1-301	Add
Zubair	64:ee:b7:c1:39:3b	ClassRoom_B1-301	Add

(b)



ID	Student Name	SSID	macAddress	Phone	Intake	Section	Action
18192103030	mh rakib	Redmi_8A	76:63:36:dfd5:fb	525892	42	1	Add Course Edit Delete
18193103007	meherab	POINT-1	6a:7ca3:90:0:af	7398432	42	1	Add Course Edit Delete
18193103034	Trisha	Trisha	36:c5:5b:21:1d:72	852	42	1	Add Course Edit Delete
18193103008	ripon	Mi_10T_Pro	7e:85:43:af:1d:28	7398432	42	1	Add Course Edit Delete
18193103001	shifat	Sifat	5c:62:8b:df:80:4	0123	42	1	Add Course Edit Delete

(c)

Fig. 6 Examples of *SmartPresence* interfaces are **a** Class Routine Search, **b** Scan active Wi-Fi, **c** Students Details, **d** Class Routine Add, **e** Take Attendance, **f** Attendance Status Report

Class Routine

Start Time:

--:-- --

End Time:

--:-- --

Subject Code:

CS101

Intake:

#Intake NUMBER

Section:

#Section NUMBER

Class Room/Device Name:

#ClassRoom_B1-301

Day:

Sunday

Update Routine

View Routine

(d)

Home Record Logout

Room Number: Enter Room Number

Course: Enter course

No Record Found

ID	Student Name	SSID	Course	Action	Active
No Record Found					

(e)

Attendance Status

Home Record Students Details Logout

Course

Intake

Section

Start Date: 12/18/2023

End Date: 12/18/2023

Search

Showing results for Course: , Intake: , Section: , Date Range: 2023-12-18 to 2023-12-18

Student ID	Name	Student Course	Time Date	DeviceMac
No Record Found				

(f)

Fig. 6 continued

The “View Routine” module empowers users to search for routines based on various parameters, implementing a search form and displaying the routine in a day-wise manner. A user-friendly interface with columns for start-time, end-time, subject code, and more adds to its functionality. The “Add Routine” module facilitates the addition

of new routine information, featuring input fields and dropdowns for an enhanced user experience. It allows for the updating of existing routine information.

The “Scan” module incorporates a reload button for deleting scanning records, providing real-time feedback, and displaying records in a visually appealing table. The success of the Classroom Scanning Application lies in its dynamic features, visually appealing interface, and user-friendly navigation.

The “Attendance Record” module establishes the foundation for a user-centric attendance management system, presenting attendance records in a tabular format for clarity. The “Attendance Status” feature enables users to input search criteria, dynamically fetch information from the database, and display results with a status bar for contextual information.

The “Student Attendance Report Card” page offers detailed attendance history, including specific dates of attendance. A “comprehensive summary card” consolidates vital information about the student, such as name, course, total days, attendance percentage, and more. This page serves as a powerful tool for assessing and understanding student attendance.

The “Course Attendance Record” module provides an overall attendance summary, including the total number of students, total attended days, and overall attendance percentage. It conditionally displays information based on attendance data availability for individual students and the overall course.

The “Student Details” module offers a clean and responsive interface for managing student details, featuring navigation buttons for easy access to different sections. It displays student information and incorporates functionality like adding courses, editing, and deleting student records.

The “Add Students” module enables the dynamic creation of student forms for adding or updating student information, ensuring a seamless and user-friendly experience. Overall, this attendance management system encompasses a holistic approach to attendance tracking, catering to the diverse needs of educators, students, and administrators.

Figure 7 shows example code snippets of our developed system.

Impact of SmartPresence

This SmartPresence; Wi-Fi-based Attendance Management System represents a groundbreaking leap in the realm of attendance tracking, ushering in a new era of efficiency, transparency, and adaptability. At the core of its impact lies the revolutionary shift from traditional, labor-intensive attendance methods to a sophisticated, automated system. By seamlessly integrating with existing educational infrastructures, this system introduces a level of scalability that spans across educational institutions, from elementary schools to universities, ensuring a comprehensive solution tailored to diverse settings. The system’s automation not only streamlines the laborious task of manual attendance tracking but also introduces real-time tracking capabilities. Leveraging mobile technology and Wi-Fi signals, the system captures attendance data with unparalleled accuracy, offering educators immediate insights into attendance patterns and enabling timely interventions. In the realm of research, the system’s data-driven approach becomes a valuable asset. Researchers can delve into granular attendance data, conducting studies on intricate aspects such as the impact of real-time tracking on student engagement, academic

```

41  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
42    Serial.println(F("SSD1306 allocation failed"));
43    for (;;)
44  }
45
46
47  pinMode(23, INPUT_PULLUP);
48  pinMode(buttonPin, INPUT);
49  // WiFi.mode(WIFI_STA); // explicitly set mode, esp defaults to STA+AP
50  // it is a good practice to make sure your code sets wifi mode how you want it.
51
52  // put your setup code here, to run once:
53  Serial.begin(115200);
54
55  //WiFiManager, local initialization. Once its business is done, there is no need
56  WiFiManager wm;
57
58  bool res;
59  res = wm.autoConnect(scannerName, ""); // password protected ap
60
61  Serial.println("Connecting to WiFi...");
62  display.clearDisplay();
63  display.setTextSize(1);
64  display.setTextColor(WHITE);
65  display.setCursor(0, 10);
66  display.println("Connecting to WiFi...");
67  display.display();
68
69  if (!res) {
70    Serial.println("Failed to connect");
71    display.clearDisplay();
72    display.setTextSize(1);
73    display.setTextColor(WHITE);
74    display.setCursor(0, 10);
75    display.println("Failed to connect !");
76    display.display();
77    // ESP.restart();
78  }
79

```

(a)

```

<table class="table table-bordered table-striped">
  <thead>
    <tr>
      <th>ssid</th>
      <th>macAddress</th>
      <th>Class Room</th>
      <th>Action</th>
    </tr>
  </thead>
  <tbody class="studentdata">
    <?php
      $query = "SELECT * FROM scan order by access_time desc,Id desc";
      $query_run = mysqli_query($con, $query);
      if(mysqli_num_rows($query_run) > 0)
      {
        foreach($query_run as $student)
        {
          <tr>
            <td><?=$student['ssid']; ?></td>
            <td><?=$student['mac']; ?></td>
            <td><?=$student['scanner']; ?></td>
            <td>
              <a href="student-create.php?macid=<?=$student['mac'];&ssid=$student['ssid'];
              ?>" target="_blank" class="btn btn-success btn-sm">Add</a>
            </td>
          </tr>
        }
      }
      else
      {
        echo "<h5> No Record Found </h5>";
      }
    <?php
  >

```

(b)

Fig. 7 Example of Code Snippets- **a** C++ code for ESP32, **b** PHP-MySQL code for Web System

performance, and the effectiveness of interventions based on attendance patterns. The comprehensive registration process, facilitated by authorized personnel, ensures that student data entered into the system is not only accurate but also securely transmitted to the central server, forming a robust foundation for data analysis. The potential research

questions spawned by the system extend to exploring its adaptability in various educational settings, uncovering the nuanced factors influencing absenteeism, and understanding the correlations between attendance records and academic outcomes. Beyond academia, this system can hold commercial promise. Its adaptability for employee attendance tracking in business settings introduces a streamlined solution for human resource management. The convenience of the system, working seamlessly on multiple platforms and offering a user-friendly interface, makes it an attractive option for businesses seeking efficient attendance-tracking solutions. The secure housing of data on an Azure cloud server addresses concerns about data security, adding a layer of trustworthiness crucial for commercial applications. One of the standout features that contribute to the system's significance is its accessibility. The web-based platform allows 24/7 access to attendance information. Human resource managers, in commercial settings, can retrieve attendance data at any time, eliminating the need to sift through physical documents. This accessibility enhances operational efficiency, providing timely information without being bound by the constraints of traditional record-keeping. The system's impact on daily practice within educational institutions is transformative. It streamlines attendance management, saving educators considerable time and reducing administrative burdens. With the ability to promptly address attendance issues through real-time data, educators can redirect their focus to the core of their profession: teaching. Moreover, the transparent access to attendance records benefits not only educators but also students and parents, fostering a collaborative educational environment. Students gain insights into their attendance history, encouraging a sense of responsibility, while parents stay informed about their child's academic progress. In the commercial realm, the system introduces features like overtime observance and location tracking. Automated calculations simplify HR tasks, allowing for accurate computation of work hours and facilitating streamlined salary processes. The location tracking feature provides real-time insights into the whereabouts of employees, ensuring accountability and notifying authorities in case of attendance anomalies.

Conclusion

The SmartPresence system revolutionizes attendance tracking, addressing the challenges of manual methods. Automating the process reduces educators' and administrators' time and effort. Real-time tracking, utilizing mobile tech and Wi-Fi, provides instant and accurate attendance data for timely interventions. Resource optimization eliminates manual data entry and paper use, promoting sustainability. Enhanced communication through web-based access fosters collaboration among educators, students, and parents. Future integration possibilities, like IoT and blockchain, ensure system adaptability to technological trends, emphasizing its significance in education.

Author contributions

Mijanur Rahman conceptualize the research and design the methodology. Md. Masudul Islam helped in conceptualization and writing of the original draft. Dip Nandi contributed to reviewing the original draft and validation the experiment. Mijanur Rahman and Md. Masudul Islam both worked for visualization, supervision, review, and editing.

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Availability of data and materials

Permanent link to code/repository used for this code version: <https://github.com/abrahametry/SmartWifiAttendance>

Declarations

Ethics approval and consent to participate.

Not applicable.

Consent for publication

Not applicable.

Competing interests

They have no conflict of interest.

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